Module 2 Quiz

Members

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Instructions: Write a Python program for each of the following problems. Apply the programming concepts covered in the corresponding sections. Each program must accept the required inputs, perform the necessary operations, and display the results.

I. Simple Formulas

1. Excavation Volume and Cost Estimation

Write a Python program that estimates the volume and total cost of soil excavation for a rectangular foundation pit. The program shall accept the length, width, and depth of the excavation in meters, along with the excavation cost per cubic meter. Compute and display the total volume of soil to be excavated and the corresponding excavation cost using the following formulas:

```
Volume = Length * Width * Depth
Total Cost = Volume * Cost per Cubic Meter
```

II. Conditions

1. Concrete Compressive Strength Classification

Write a Python program that classifies concrete based on its compressive strength. The program shall accept the compressive strength in megapascals (MPa) and display the corresponding classification according to the following simplified scale:

Compressive Strength	Classification
≥ 50 MPa	Very High Strength
≥ 40 MPa	High Strength
≥ 30 MPa	Moderately High Strength
≥ 20 MPa	Normal Strength
< 20 MPa	Low Strength

For this exercise, use the given classification thresholds regardless of other engineering standards.

III. Conditions and Formulas

1. Cross-Sectional Area of Structural Members

Write a Python program that calculates the cross-sectional area of a selected structural member. The program shall allow the user to choose among a rectangular, circular, triangular, or hollow rectangular cross-section. Depending on the selected shape, the program shall request the necessary dimensions in millimeters and compute the area using the appropriate formula below. Display the result in square millimeters.

```
Rectangular = Width * Height
Circular = 3.1416 * Radius ** 2
Triangular = (Base * Height) / 2
Hollow Rectangular = (Outer Width * Outer Height)
                   - (Inner Width * Inner Height)
```

IV. Loops

1. Construction Material Code Validation and Formatting

Write a Python program that accepts a construction material identification code containing letters, numbers, and optional dashes. The program shall validate that the code contains exactly 10 alphanumeric characters, excluding dashes, and that no other special characters are present. Using a loop, replace all alphanumeric characters with asterisks (*) except the last three, while preserving any existing dashes. Display an appropriate error message for invalid inputs.

Examples:

```
Input:  CEM-2026-AB7
Output: ***-****-AB7

Input:  STEEL12345
Output:  *******345

Input:  CEM@2026AB7
Output: Invalid material code
```